ASSIGNMENT No: 01 (BI)
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Title: DNA Sequence Analysis Problem Statement: Analyze a given DNA sequence and perform basic sequence manipulation, including finding motifs, calculating GC content, and identifying coding regions.
Code:

```
dna="ATGCGATCGATCGATGCTAGCTAGCTAACGATGAAATGCGTAGCTAGCTAA"
dna=dna.upper()

print("DNA SEQUENCE ANALYSIS")
print("-"*40)
print("DNA Sequence:",dna)
print("Sequence Length:",len(dna))

print("\n1. BASIC SEQUENCE MANIPULATION")
print("-"*40)
print("Number of A:",dna.count("A"))
print("Number of T:",dna.count("T"))
print("Number of G:",dna.count("G"))
print("Number of C:",dna.count("C"))

complement=dna.translate(str.maketrans("ATGC","TACG"))
reverse=dna[::-1]
reverse_complement=complement[::-1]

print("Complement:",complement)
print("Reverse:",reverse)
print("Reverse Complement:",reverse_complement)

print("\n2. MOTIF IDENTIFICATION")
print("-"*40)
motif="ATG"
print("Motif:",motif)

positions=[]
start=0
while True:
    position=dna.find(motif,start)
    if position==-1:
        break
    positions.append(position+1)
    start=position+1

if positions:
    print("Motif found at positions:",positions)
```

```
    print("Number of occurrences:",len(positions))
else:
    print("Motif not found.")

print("\n3. GC CONTENT")
print("-"*40)
g_count=dna.count("G")
c_count=dna.count("C")
gc_content=((g_count+c_count)/len(dna))*100

print("Number of G:",g_count)
print("Number of C:",c_count)
print("GC Content: {:.2f}%".format(gc_content))

print("\n4. CODING REGION IDENTIFICATION")
print("-"*40)
start_codon="ATG"
stop_codons=["TAA","TAG","TGA"]
orfs=[]

for i in range(len(dna)-2):
    if dna[i:i+3]==start_codon:
        for j in range(i+3,len(dna)-2,3):
            codon=dna[j:j+3]
            if codon in stop_codons:
                orf=dna[i:j+3]
                orfs.append((i+1,j+3,orf))
                break

if orfs:
    print("Potential Coding Regions / ORFs:")
    for number,(start,end,sequence) in enumerate(orfs,1):
        print("\nORF",number)
        print("Start Position:",start)
        print("End Position:",end)
        print("Length:",len(sequence))
        print("Sequence:",sequence)
else:
    print("No potential coding regions found.")

print("\nAnalysis completed successfully.")
print("Aditya Rade BEAD23167")
```

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OUTPUT:

```
ORF 4
Start Position: 36
End Position: 47
Length: 12
Sequence: ATGCGTAGCTAG

Analysis completed successfully.
Aditya Rade   BEAD23167
PS C:\Users\adity\OneDrive\Documents\Desktop\Sem 7 study materials\CL - 2> ^T^T^T
```

Problems 10 Debug Console Output Terminal Ports Jupyter

```
PS C:\Users\adity\OneDrive\Documents\Desktop\Sem 7 study materials\CL - 2> & C:\Users\adity\AppData\Loc
OneDrive/Documents/Desktop/Sem 7 study materials/CL - 2/Practical-1-BI.py"
Number of occurrences: 4

3. GC CONTENT
----------------------------------------
Number of G: 13
Number of C: 10
GC Content: 45.10%

4. CODING REGION IDENTIFICATION
----------------------------------------
Potential Coding Regions / ORFs:

ORF 1
Start Position: 1
End Position: 24
Length: 24
Sequence: ATGCGATCGATCGATGCTAGCTAG

ORF 2
Start Position: 14
End Position: 28
Length: 15
Sequence: ATGCTAGCTAGCTAA

ORF 3
Start Position: 31
End Position: 51
Length: 21
Sequence: ATGAAATGCGTAGCTAGCTAA
```